Linear models

1 Some friends spent a total of $\$5.00$ on stickers and badges at a store. A sheet of stickers costs $\$0.50$ and a badge costs $\$1.50$.

a Write a linear equation that represents the relationship between the number of sheets of stickers, x, and the number of badges, y, the friends bought for $\$5.00$.

b If the friends bought 2 badges, how many sheets of stickers did they buy?

2 The cost of a taxi ride is given by $C = 5.5t + 3$, where t is the duration of the trip in minutes.

a Calculate the cost of an 11-minute trip.

b For every extra minute the trip takes, how much more will the trip cost?

c What could the constant value of 3 represent in context?

3 A baseball is thrown vertically upward by a baseball player when he is standing on the ground, and the velocity of the baseball, V (in yards per second), after T seconds is given by $V = 120 - 32T$.

a Complete the table of values:

Time	0	1	2	3	4
Vertical velocity					

b State the slope of the linear function.

c Explain the negative value of V when $T = 4$.

4 A carpenter charges a callout fee of $\$150$ plus $\$45$ per hour.

a Write a linear equation to represent the total amount charged, y, by the carpenter as a function of the number of hours worked, x.

b State the slope of the linear function.

c Describe what the slope of the line represents in this context.

d State the y-intercept.

e Describe what the y-intercept represents in this context.

f Find the total amount charged by the carpenter for 6 hours of work.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

5 Mohamad is taking his new Subaru out for a drive. He had only driven 50 miles in it before and is now driving it down the highway at $75\ \text{mi/h}$.

a Write a linear equation to represent the total distance, y, that Mohamad had driven in his Subaru as a function of the number of hours, x.

b State the slope of the function.

c Describe what the slope of the line represents in this context.

d State the y-intercept.

e Describe what the y-intercept represents in this context.

f Find the total distance Mohamad will have driven in his Subaru if his current drive begins at 5:10 pm and finishes at 7:25 pm.

6 The table shows the linear relationship between the length of a cellphone call and the cost of the call:

Length of call in minutes (x)	1	2	3
Cost in dollars (y)	7.6	14.4	21.2

a Write a linear equation to represent the cost of a call, y, as a function of the length of the call, x.

b State the slope of the function.

c Describe what the slope represents in this context.

d State the y-intercept.

e Describe what the y-intercept could represent in this context.

f Find the cost of a 6-minute call.

7 Mario is running a $100\ \text{km}$ ultramarathon at an average speed of $9\ \text{km/h}$.

a Write a linear equation to represent the distance Mario has left to run, y, as a function of the number of hours since the start, x.

b State the slope of the function.

c Describe what the slope of the line represents in this context.

d State the y-intercept.

e Describe what the y-intercept represents in this context.

f Find the distance Mario will have left to run after 4.5 hours.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 A particular restaurant has a fixed weekly cost of $\$1300$ and receives an average of $\$16$ from each customer.

- **a** Write an equation to represent the net profit, y, of the restaurant for the week as a function of the number of customers, x.
- **b** Find the slope of the function.
- **c** Describe what the slope of the line represents in context.
- **d** Find of the y-intercept.
- **e** Describe what the y-intercept represents in context.
- **f** Find the restaurant's net profit if it has 310 customers for the week.

9 A cellphone salesman earned $\$600$ in a particular week during which he sold 26 phones and $\$540$ in another week during which he sold 20 phones.

- **a** Write a linear equation to represent the weekly earnings of the salesman, y, as a function of the number of phones sold, x.
- **b** State the slope of this function.
- **c** Describe what the slope of the line represents in context.
- **d** Find the y-intercept.
- **e** Describe what the y-intercept represents in context.
- **f** Find how much the salesman will earn in a week during which he sells 36 phones.

10 In a study, scientists found that the more someone sleeps, the quicker their reaction time. The table below displays the findings:

Number of hours of sleep (x)	0	1	2	3	4	5
Reaction time in seconds (y)	6	5.8	5.6	5.4	5.2	5

- **a** How much does the reaction time decrease for each extra hour of sleep?
- **b** Write a linear equation relating the number of hours of sleep, x, and the reaction time, y.
- **c** Calculate the reaction time for someone who has slept 4.5 hours.
- **d** Calculate the number of hours someone sleeps if they have a reaction time of 5.5 seconds.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 A diver starts at the surface of the water and begins to descend below the surface at a constant rate. The table below shows the depth of the diver, in yards, over 4 minutes:

Number of minutes passed (x)	0	1	2	3	4
Depth of diver in yards (y)	0	1.4	2.8	4.2	5.6

a Calculate the increase in depth each minute.

b Write a linear equation for the relationship between the number of minutes passed, x, and the depth, y, of the diver.

c Calculate the depth of the diver after 6 minutes.

d Calculate how long the diver takes to reach 12.6 yd beneath the surface.

12 After Mae starts running, her heartbeat increases at a constant rate.

a Complete the following table:

Number of minutes passed (x)	0	2	4	6	8	10	12
Heart rate (y)	49	55	61	67	73	79	

b What is the unit change in y for the above table?

c Write a linear equation that describes the relationship between the number of minutes passed, x, and Mae's heartbeat, y.

d Describe what the slope represents in this context.

13 Paul has just purchased a prepaid phone, which he intends to use exclusively for sending text messages, and has purchased some credit along with it to use. After sending 11 text messages, he has $\$34.39$ of credit remaining and after sending 19 text messages, he has $\$30.31$ of credit remaining.

a The relationship between the number of text messages sent and the amount of credit remaining is linear. Determine the slope of the linear function.

b Write a linear equation to represent the amount of credit remaining, y, as a function of the number of text messages sent, x.

c Describe what the slope of the line represents in this context.

d State the value of the y-intercept.

e Describe what the y-intercept represents in this context.

f Find how much credit Paul will have left after sending 36 text messages.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

14 A racing car starts the race with $150\ \text{L}$ of fuel. From there, it uses fuel at a rate of $5\ \text{L}$ per minute.

a Complete the following table of values:

Number of minutes passed (x)	0	5	10	15	20
Amount of fuel left in the tank (y)					

b Write a linear equation linking the number of minutes passed, x, and the amount of fuel left in the tank, y.

c Explain the meaning of the slope in this context.

d How many minutes will it take for the car to run out of fuel?

15 The table shows the linear relationship between the number of plastic chairs manufactured and the total manufacturing cost:

Number of plastic chairs (x)	2	4	7
Cost in dollars (y)	135	185	260

a Write a linear equation to represent the total manufacturing cost, y, as a function of the number of plastic chairs manufactured, x.

b State the slope of the function.

c Describe what the slope of the function represents in this context.

d State the y-intercept.

e Describe what the y-intercept could represent in this context.

f Find the total cost of manufacturing 13 plastic chairs.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

16 The table shows the water level of a well that is being emptied at a constant rate with a pump:

Time in minutes (x)	2	5	8
Water level in meters (y)	26.8	25	23.2

a Write a linear equation to represent the water level, y, as a function of the minutes passed, x.

b State the slope of the function.

c Describe what the slope of the function represents in this context.

d State the y-intercept.

e Describe what the y-intercept represents in this context.

f Calculate the water level be after 15 minutes.

Graphs of linear models

17 There are 20 L of water in a rainwater tank. It rains for a period of 24 hours and during this time, the tank fills up at a rate of 8 L per hour.

a Complete the table of values:

Number of hours passed (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)							

b Write a linear equation linking the number of hours passed, x, and the amount of water in the tank, y.

c Plot the points on a coordinate plane.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18 Consider the points in the following table:

Time in minutes (x)	1	2	3	4	5
Temperature in $^\circ$F (y)	8	13	18	23	28

a By how much does the temperature increase each minute?

b What would the temperature have been at time 0?

c Write a linear equation that represents the relationship between x and y.

d Sketch the graph of the linear relationship.

19 A carpenter charges a callout fee of $\$90$ plus $\$30$ per hour.

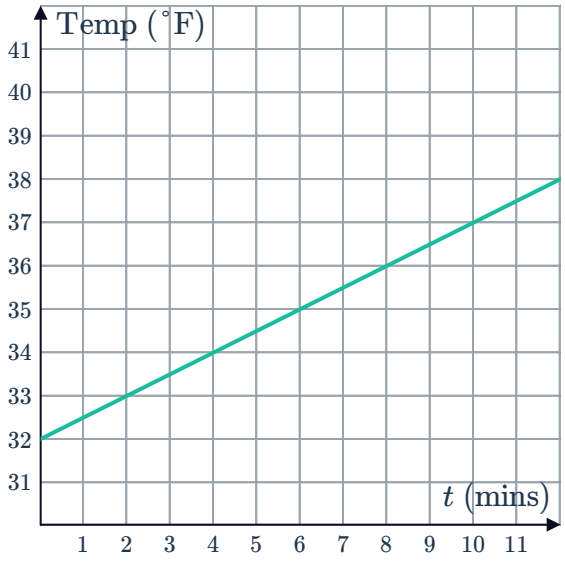
a Write a linear equation to represent the total fee charged, y, by the carpenter as a function of the number of hours worked, x.

b Complete the table of values:

Number of hours worked (x)	0	1	2	3	4	4.5	7
Total fee charged (y)							

c Sketch the graph of the linear relationship.

20 The graph shows the temperature of a room after the heater has been turned on:



a State the slope of the line.

b State the y-intercept.

c Write a linear equation to represent the temperature of the room, y, as a function of time, t.

d Describe what the slope represents in this context.

e Describe what the y-intercept represents in this context.

f Find the temperature of the room after the heater has been turned on for 40 minutes.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

21 A ball is rolled down a hill. The table below shows the velocity, V, of the ball after a given number of seconds, t:

Time in seconds (t)	0	1	2	3	4	5
Velocity in m/s (V)	12	13.3	14.6	15.9	17.2	18.5

a Sketch a graph that displays the ball's velocity against time.

b Calculate the slope of the line.

c Describe what the slope represents in this context.

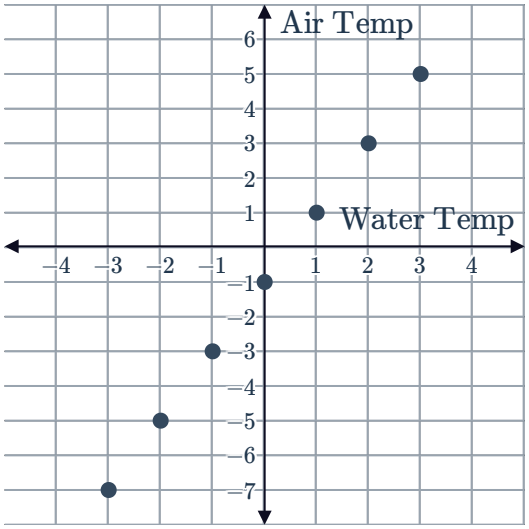
d State the y-intercept of the line.

e Describe what the y-intercept represents in this context.

f Write a linear equation for the line, expressing V in terms of t.

g Now, determine the velocity of the ball after 19 seconds.

22 The following graph shows the relationship between water temperatures and surface air temperatures:



a Complete the table of values:

Water temperature (°F)	−3	−2	−1	0	1	2	3
Surface air temperature (°F)							

b Write a linear equation representing the relationship between the water temperature, x, and the surface air temperature, y.

c Find the surface air temperature when the water temperature is 14°F.

d Find the water temperature when the surface air temperature is 23°F.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

23 It starts raining and an empty rainwater tank fills up at a constant rate of $2\ \text{gal}$ per hour. By midnight, there are $20\ \text{gal}$ of water in the rainwater tank. As it rains, the tank continues to fill up at this rate.

a Complete the table of values:

Number of hours passed since midnight (x)	0	1	2	3	4	4.5	10
Amount of water in tank (y)							

b Sketch the graph depicting the situation on a coordinate plane.

c Write a linear equation linking the number of hours passed since midnight, x, and the amount of water in the tank, y.

d Determine the y-intercept of the line.

e At what time prior to midnight was the tank empty?

24 The number of fish in a river is approximated over a five year period. The results are shown in the following table:

Time in years (t)	0	1	2	3	4	5
Number of fish (F)	4800	4600	4400	4200	4000	3800

a Sketch a graph corresponding to the given information.

b Calculate the slope of the line.

c Describe what the slope represents in this context.

d State the value of F when the line crosses y-axis.

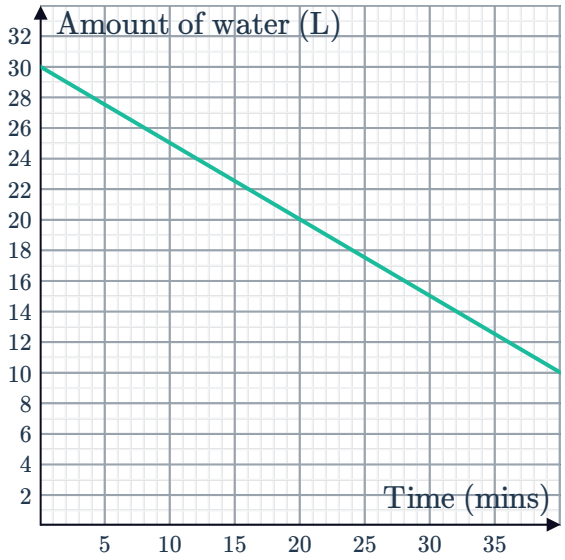
e Determine an equation for the line using the given values.

f Now, determine the number of fish remaining in the river after 13 years.

g Find the number of years, t, until 2000 fish remain in the river.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

25 The graph shows the amount of water remaining in a bucket that was initially full before a hole was made in it's side:



a State the slope of the line.

b Find the y-intercept.

c Write a linear equation to represent the amount of water remaining in the bucket, y, as a function of time, x.

d Describe the meaning of the slope in this context.

e Describe the meaning of the y-intercept in this context.

f Find the amount of water remaining in the bucket after 54 minutes.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647